

SCIENCE INVESTIGATIONS
PLANNING AND REPORT WORKSHEET

NAME: SOLUTIONS - TEMPERATURE + SOLUBILITY.

CLASS: 20

OTHER GROUP MEMBERS: _____

Aim

- What are you going to investigate?

The effect of temperature on the solubility of a substance. (1)

Hypothesis

- What do you think will happen? Explain why.

Equipment

- List the equipment you will need.

Test tube

10 mL graduated cylinder

popstick

thermometer

250 mL beaker

electric kettle (or other heating device).

solute (CuSO_4 , NH_4Cl , salt, sugar, KNO_3 , etc).

(1)

Method

- Which variable(s) are you going to

change? (independent)

temperature of distilled water.

(1)

measure? (dependent)

amount of solute dissolved.

(1)

keep the same? (controlled)

same size
"popstickfuls".

volume of distilled water.

size of test tube.

same thermometer.

(Any 2 - 1 mark each).

- How will you make it a fair test?

- List the steps you will take to do the investigation and collect the results. Use labelled diagrams if appropriate.

Suitable diagram with labels. (1)

Steps in logical sequence. (2)

At least 2 trials (1).

Results

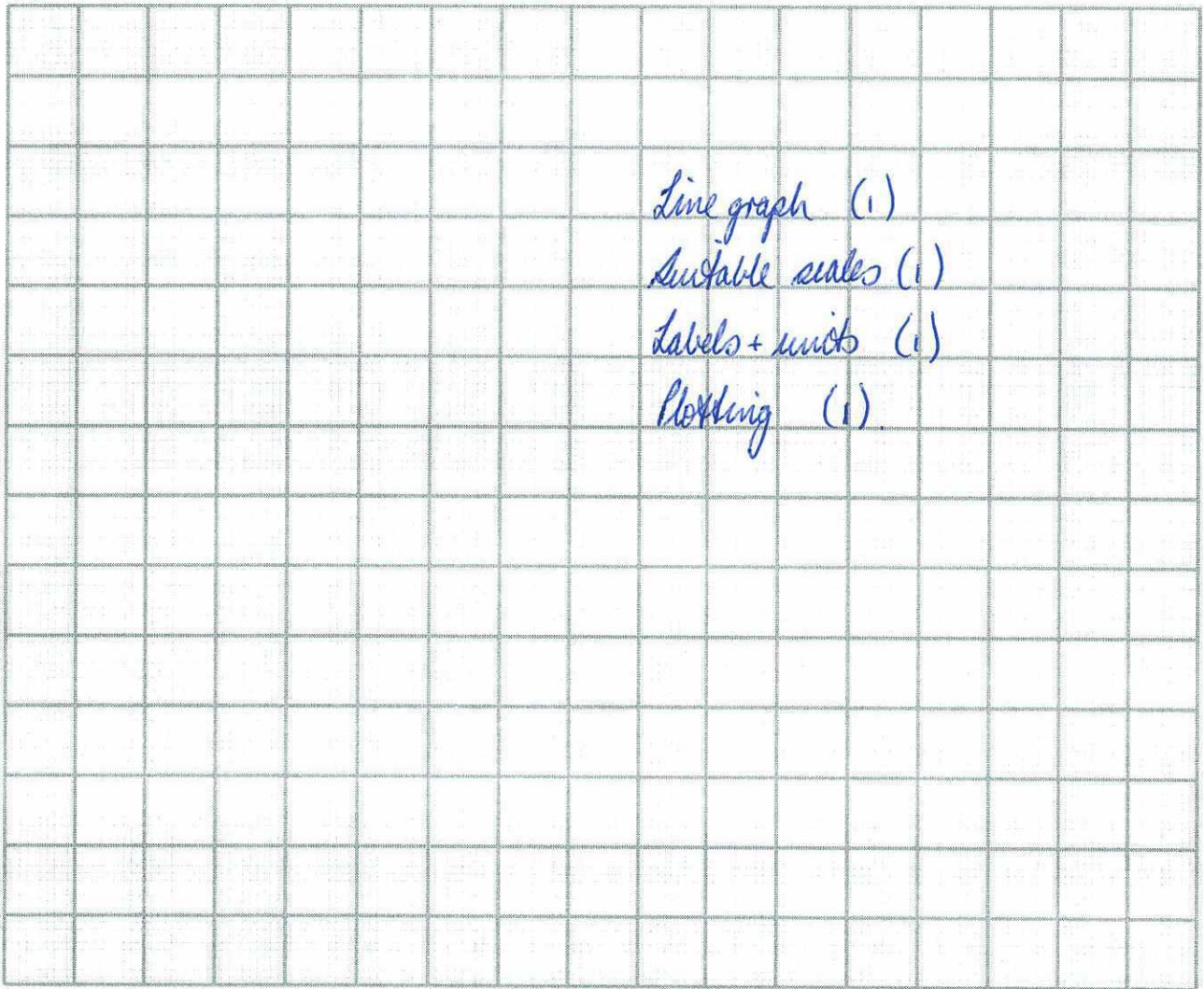
- Record your results or observations. Use a table if appropriate.

Must be a table.

Headings + units (1)

Clearly set out. (1).

- Can your results be represented in a graph?



**Discussion + •
 Critical Analysis**

What do your results tell you? Are there any patterns or trends in your results?

Adequately describes a shape or trend. (1)
 (Depends - some solutes DECREASE solubility with temp.)

- Using some Science ideas, can you explain why you got these patterns or trends?

Makes a reasonable attempt at explaining the observations. (1)

- What things went wrong or affected the results you obtained? Suggest improvements for this investigation if possible.

Makes at least one suggestion for improvement or notes at least one problem with the technique used (1)

Conclusion

- What is the relationship you found from this investigation? It should answer the question asked in the "aim".

Should state whether solubility increases or decreases. (1)